

PAVAN SAI CHILUKALA

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PROFESSIONAL SUMMARY

Full-stack and mobile software engineer with professional experience building and shipping web, mobile, and AI-integrated applications. Currently building an Agentic Financial OS at Harmony Financial Technologies using React, Node.js, Supabase, Stripe, Plaid, and Auth0. Proficient in TypeScript, JavaScript, Swift, Python, Java, GraphQL, PostgreSQL, Neo4j, MongoDB, Docker, AWS, and CI/CD pipelines. M.S. Computer Science (GPA 3.8, December 2025). Actively uses Claude Code, Cursor, and GitHub Copilot daily. On OPT, open to relocate anywhere in the US. Available immediately.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Swift, Python, Java, C/C++, PHP, SQL
Mobile: React Native, SwiftUI, Expo, iOS (App Store), Android (Play Store), Core Audio (CATap)
Frontend: React, Next.js, Vite, Redux, Tailwind CSS, Bootstrap, HTML5/CSS3
Backend: Node.js, Express.js, GraphQL, Flask, FastAPI, REST API Design
Auth & Payments: Auth0, Supabase, Stripe, Plaid, JWT, Google OAuth
Databases: PostgreSQL, MySQL, Neo4j, MongoDB, Cassandra, Firebase, Firestore
AI / ML: OpenAI API, Anthropic Claude API, LLM Integration, Prompt Engineering, PyTorch, NLP, LangGraph
Visualization: D3.js, Plotly, Matplotlib, Tableau
Cloud & DevOps: AWS S3, AWS EC2, AWS Lambda, AWS ECS, AWS DynamoDB, AWS SQS, GCP, Docker, Kubernetes, CI/CD, GitHub Actions, Vercel, Railway, Render
AI Dev Tools: Claude Code, GitHub Copilot, Cursor, ChatGPT, Gemini
Tools: Git, GitHub, Postman, Xcode, VSCode, PyCharm, WebStorm, Linux (Ubuntu)

WORK EXPERIENCE

Software Engineer

May 2026 – Present
Remote

Harmony Financial Technologies

- Building an Agentic Financial OS with KYA (Know Your Agent) protocol – infrastructure enabling AI agents to initiate and settle payments between companies autonomously.
- Full-stack development in React and Node.js with Supabase backend, Stripe and Plaid for payment processing and financial data aggregation, Auth0 for multi-tenant agent identity management, deployed on Railway and Vercel.
- Designing agentic workflow orchestration for cross-company financial transactions at a fast-moving startup with full ownership and minimal direction.

iOS Developer Volunteer

Feb 2026 – Jun 2026
Remote

Streetcare

- Built and shipped new features and resolved bugs in a production Swift and SwiftUI iOS application used by field workers supporting homeless communities across the US.
- Integrated Firestore for real-time cloud sync and CoreData for local persistence; performed technical root cause analysis, peer code reviews, and release management via App Store Connect and TestFlight.

Software Engineer

Sep 2022 – Nov 2023
Hyderabad, India

Dhanush Healthcare Systems

- Built **60+ modular UI components** to enhance frontend scalability; applied memoization across 30+ components achieving **80% faster TAT**; used lazy async loading to achieve **70% faster UI responsiveness**.
- Integrated **150+ RESTful APIs** for real-time mobile/backend synchronization; published **16 production releases** to App Store and Play Store.
- Integrated Google OAuth and leveraged Firebase to create **35+ dynamic deep-linked (Redux) navigation flows**; managed CI/CD via GitHub Actions in a strict Git-flow environment.

Graduate Teaching Assistant – Algorithms & Data Structures

Jan 2025 – Dec 2025
Warrensburg, MO

University of Central Missouri

- Evaluated 500+ assignments for logical correctness, time complexity (Big O), and algorithmic efficiency across 50+ undergraduate students; provided structured written feedback on pseudocode and implementation quality.
- Mentored students on optimal data structure selection and scalable system design; collaborated with faculty to develop course content.

PROJECTS

volumeMixer <i>Swift, SwiftUI, Core Audio (CATap), macOS</i> github/volumeMixer	2026
– Built a macOS menu bar utility using low-level Core Audio Tap (CATap) APIs at the hardware abstraction layer with no third-party dependencies; designed from first principles and shipped as open-source.	
AI Full-Stack Platform <i>React, TypeScript, Node.js, PostgreSQL, AWS, OpenAI, Anthropic</i>	2025
– Built an end-to-end AI-integrated web application with OpenAI and Anthropic LLM APIs: streaming responses, prompt engineering, tool calls, and agentic workflows; deployed on AWS with CI/CD at 99.9% uptime .	
The Daily Wag <i>React, TypeScript, Next.js, Node.js, MySQL, AWS S3, Docker, JWT</i> github/daily-wag	2025
– Led full-stack architecture of a pet clinic consumer platform; engineered custom scheduling APIs preventing concurrent booking conflicts; built admin dashboard with analytics across 200+ products ; deployed via Docker and CI/CD with zero-downtime releases.	
C3-Carpool Connect <i>React, TypeScript, Python Flask, PostgreSQL, MongoDB</i> github/carpool-frontend	2024
– Built a real-time ride-sharing platform achieving sub-second matching latency via optimized geo-spatial queries; maintained 99.9% uptime via automated CI/CD pipelines on Vercel and Render.	
Black Friday Sales Prediction <i>Python, PyTorch, ML, NLP, D3.js, Plotly, Matplotlib, Tableau</i>	2022
– Developed a Python ML model using PyTorch and NLP to analyze customer behavior patterns; built interactive dashboards using D3.js, Plotly, Matplotlib, and Tableau.	

EDUCATION

Master of Science, Computer Science GPA: 3.8 / 4.0	<i>Jan 2024 – Dec 2025</i>
<i>University of Central Missouri</i>	<i>Warrensburg, MO</i>
Bachelor of Technology, Computer Science & Engineering GPA: 3.75 / 4.0	<i>Aug 2019 – Jul 2023</i>
<i>Jawaharlal Nehru Technological University (JNTU)</i>	<i>India</i>

CERTIFICATIONS

React Native – The Practical Guide [2023]	<i>Udemy</i>
Database Design and Basic SQL in PostgreSQL	<i>Coursera</i>
Intermediate PostgreSQL	<i>Coursera</i>
Supervised Machine Learning: Regression and Classification	<i>Coursera / DeepLearning.AI</i>